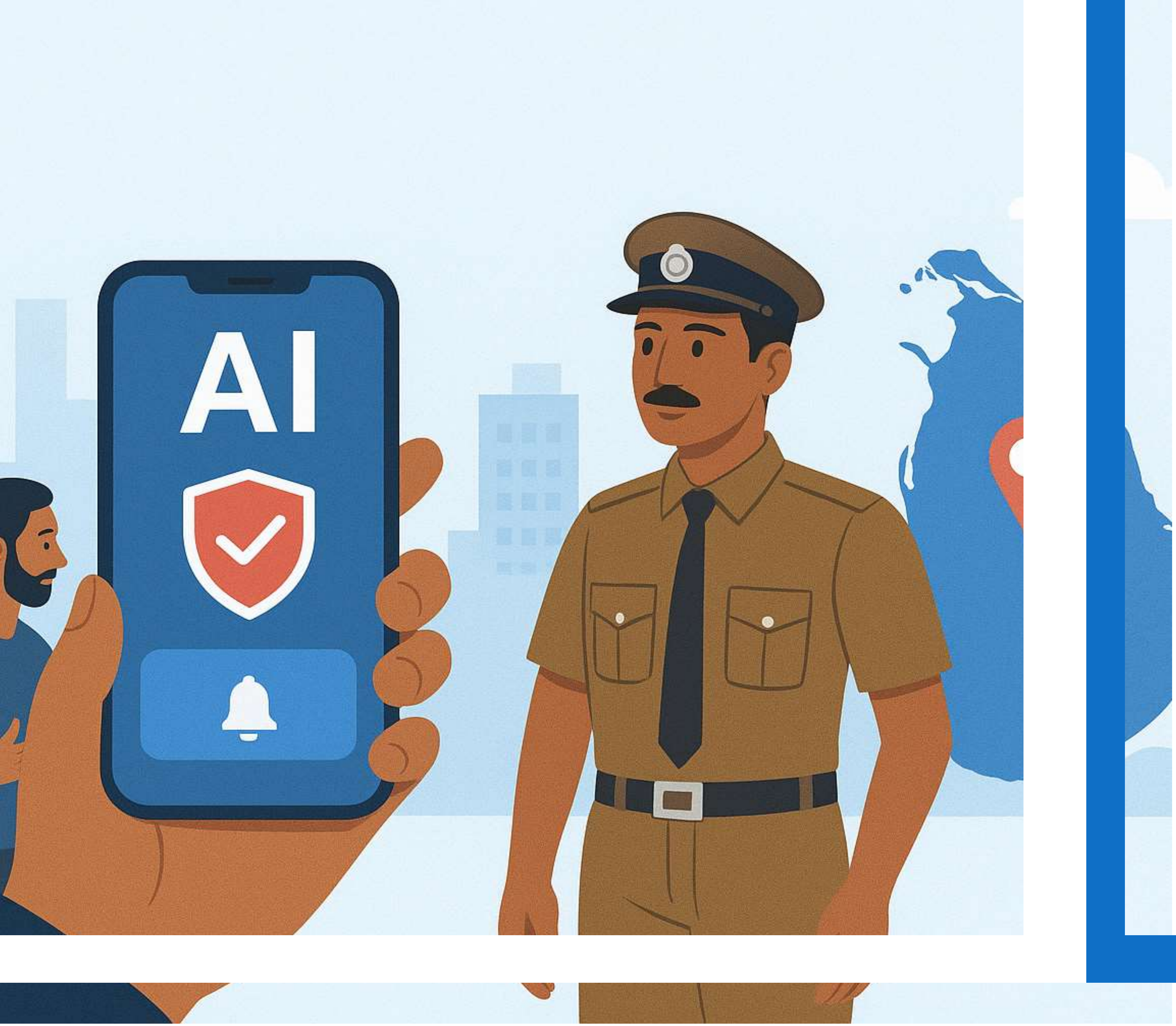


SafeLink : AI-Powered Mobile Platform Enhancing Public Safety and Community Policing in Sri Lanka

Project ID: 25-26J-200

Team

Supervisor
Ms. Jenny Krishara

Co-supervisor
Ms. Jenny Krishara

Team Leader
Indudunu I W O

Amaradasa S A A M

Fernando P.K.S

Jayakody J.M.P.S.B.

SAFEY PREDICTION FOR WOMEN AND CHILDREN

Specialization : Information Technology

Safey Prediction for Women and Children

Knowledge Gap / Problem Definition

- Mostly focus on GPS tracking or IoT devices.
- Detect emergencies but lack predictive safety insights.
- No reliable Sri Lanka–focused solution.
- Bilingual support (Sinhala-English) is missing.
- Crowdsourced inputs often untrusted or unfiltered.

Shortcomings of Existing Systems

- Reactive (alerts only after incident happens).
- Depend on manual activation (panic button, SMS).
- Some prediction models exist (Huber regression, etc.) but limited.
- Rarely apply clustering for geospatial risk zones.
- Lack trust-based filtering for community reports.

Safey Prediction for Women and Children

Research Gap Comparison

Features	[2]	[5]	[7]	[8]	DiverseMind
Incident classification (harassment, unsafe transport)	✗	✗	✗	✗	✓
Predictive analytics (crime patterns, clustering)	✗	✗	✓	✗	✓
Bilingual Sinhala–English NLP	✗	✗	✗	✗	✓
emergency alert or Safety tips	✓	✓	✗	✓	✓
Crowdsourced safety ratings (trust-based)	✗	✗	✗	✗	✓
Heatmap & risk-zone visualization	✗	✗	✓	✗	✓
Police dashboard and community integration	✗	✗	✓	✗	✓

Safey Prediction for Women and Children

Proposed Creative Solution

SafeLink

AI filtering of reports and classify women & child incidents.

Risk prediction

Heatmaps & bilingual alerts.

Crowdsourced safety ratings.

Integration with dashboards.

Safey Prediction for Women and Children

Key Domains

- Artificial Intelligent (AI)
- Natural Language Processing (NLP)
- Crowdsourcing and Trust Models
- Mobile and Cloud
- Machine Learning

Safey Prediction for Women and Children

Latest Technologies

- Sinhala-English NLP for code-switching.
- DBSCAN, ST-DBSCAN, HDBSCAN + time for spatiotemporal clı
- TensorFlow, FastAPI, Flutter, Firebase.
- GIS visualization for heatmaps.

Firebase

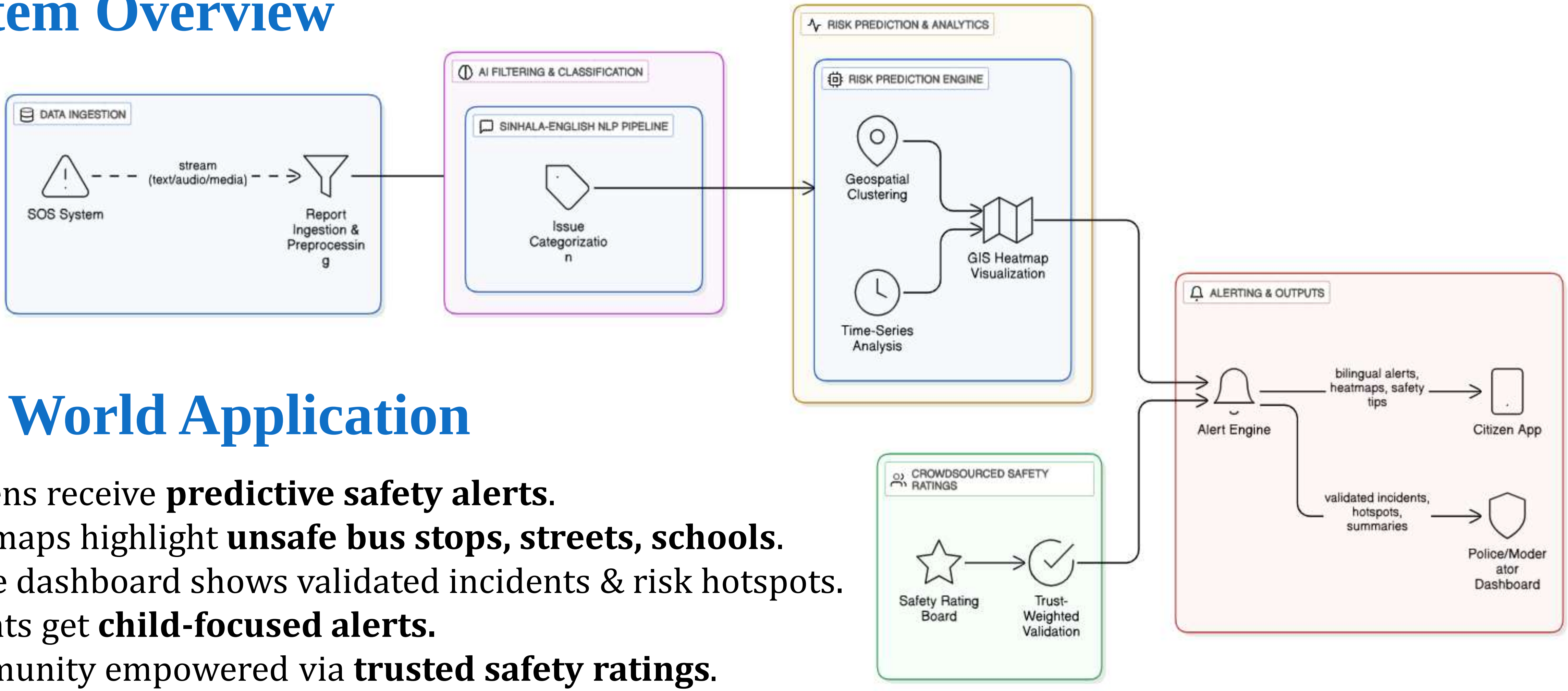
FastAPI

Data Availability and Ethical Clerance

- Classifier accuracy ≥ 85%.
- Alerts delivered in < 3s.
- Risk map validation with real records.
- Ethical clearance: anonymized, encrypted, user-consent based.
- Data: SOS reports, survey data, crowdsourced ratings.

Safey Prediction for Women and Children

System Overview

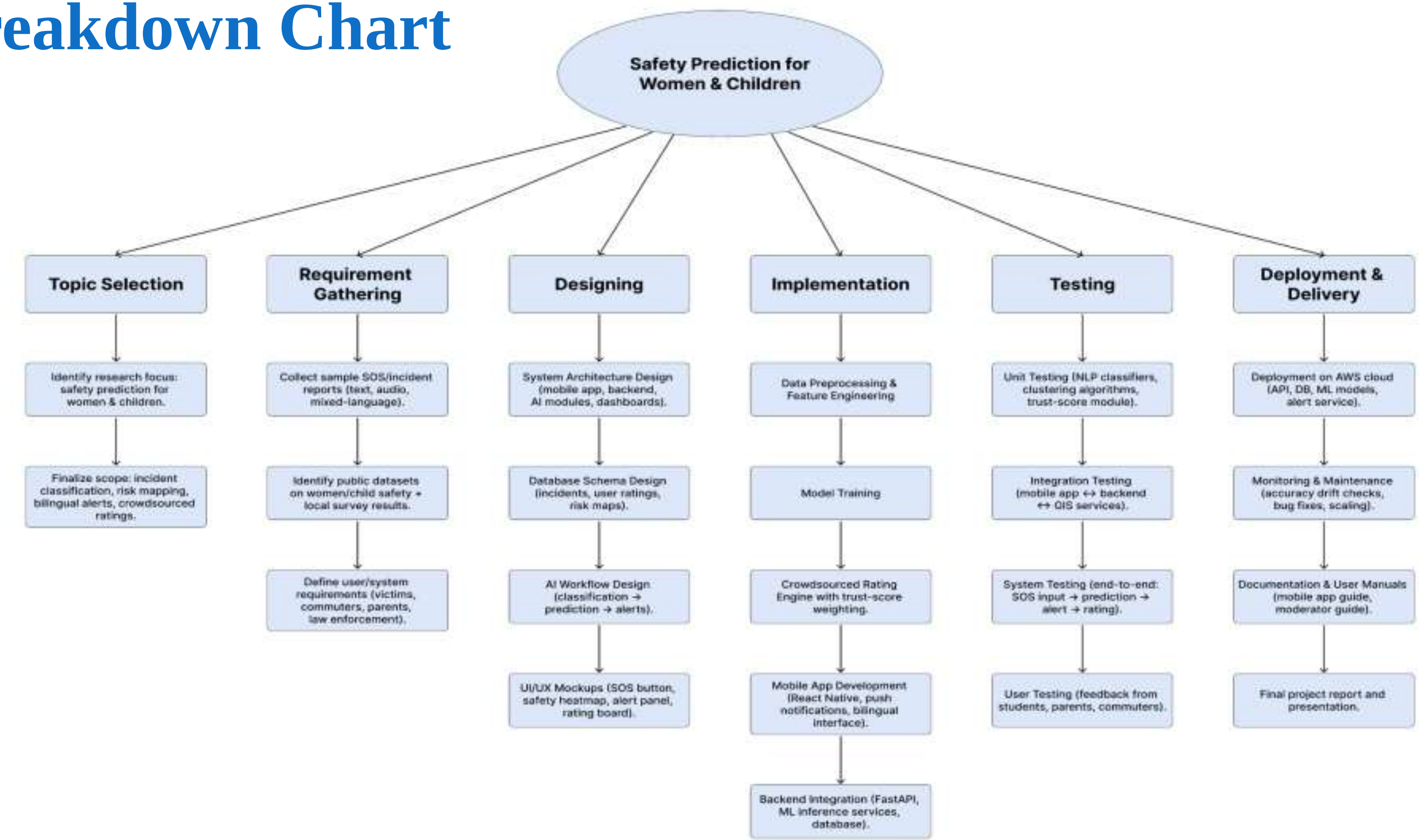


Real World Application

- Citizens receive **predictive safety alerts**.
- Heatmaps highlight **unsafe bus stops, streets, schools**.
- Police dashboard shows validated incidents & risk hotspots.
- Parents get **child-focused alerts.**
- Community empowered via **trusted safety ratings**.

Safey Prediction for Women and Children

Work Breakdown Chart



Safey Prediction for Women and Children

References

[1] A. Gupta and V. Harit, “Child Safety & Tracking Management System by using GPS, Geo-Fencing & Android Application: An Analysis,” *Proc. 2016 2nd Int. Conf. on Computational Intelligence & Communication Technology (CICT)*, 2016.

[2] A. Srinivasan, A. R., S. Abirami, D. N., and S. B. Sreeja, “Intelligent Child Safety System using Machine Learning in IoT Devices,” *Proc. 2020*.

[3] R. E., K. Chozhan, M. Jagadesh, S. Abinaya, V. S. Agilan, and N. Abinaya, “Child Safety Monitoring System,” *Proc. 2024 10th Int. Conf. on Advanced Computing and Communication Systems (ICACCS)*, 2024.

[4] K. Tamilarasi, T. Revathi, B. Madhumitha, and S. Shruthi, “Women Safety Analytics,” *Proc. 2024 IEEE Int. Conf. on System, Computation, Automation and Networking (ICSCAN)*, 2024.

[5] P. Gupta, P. Rastogi, Y. Goyal, S. Saboo, and D. Gulati, “Women’s Safety Device Based on Internet of Things,” *Proc. 2021 IEEE Int. Conf. on Mobile Networks and Wireless Communications (ICMNWC)*, 2021.

[6] A. Arundhati and G. Sinha, “Women Safety System with Emergency Alert, Analysis and Prediction,” *Proc. 2024 1st Int. Conf. on Technological Innovations and Advance Computing (TIACOMP)*, 2024.

[7] A. Avinashiappan, H. Thiagarajan, H. C. Mahesh, and R. Suresh, “Smart Surveillance using Deep Learning,” *Int. J. of Reconfigurable and Embedded Systems (IJRES)*, vol. 12, no. 3, pp. 423–432, Nov. 2023.

[8] S. K. Punjabi, D. Reddy, S. Chaure, and U. Ravale, “Smart Intelligent System for Women and Child Security,” *Proc. 2018.*

EMERGENCY SOS WITH REAL-TIME LOCATION & LIVE STREAMING

Specialization : Information Technology

Emergency SOS with Real-Time Location & Live Streaming

Knowledge Gap / Problem Definition

- Emergencies in Sri Lanka are reported mainly by voice calls (119).
- Victims often cannot speak clearly due to scared, stress, or injury.
- Existing SOS apps only send GPS or text, not the seriousness of the case.

Voice Call Limitations

Delayed Responses

Shortcomings of Existing Systems

- Voice-only reporting is unreliable in emergencies.
- GPS-only alerts cannot differentiate between severity levels.
- Need for a system with real-time video, audio, and AI detection

No AI Prioritization

Emergency SOS with Real-Time Location & Live Streaming

Research Gap

Features	[1]	[2]	[3]	[4]	SafeLink
GPS Location Sharing	✓	✓	✗	✓	✓
Live Video Streaming	✗	✗	✓	✗	✓
Live Audio Streaming	✗	✗	✗	✗	✓
AI-based Threat Detection (weapons, fire, sounds)	✗	✗	✓	✓	✓
Severity Scoring & Prioritization	✗	✗	✗	✗	✓
Police Dashboard with Real-Time Monitoring	✗	✗	✗	✗	✓
Secure Geo-Tagged Evidence Storage	✗	✗	✗	✗	✓
Tailored for Sri Lanka Context	✗	✗	✗	✗	✓

Proposed Creative Solution

SafeLink

» One-tap SOS button to trigger GPS + live streaming.

» YOLOv8 YAMNet Analysis

» Risk scoring using crime data for prioritization.

» Police dashboard for real-time monitoring.

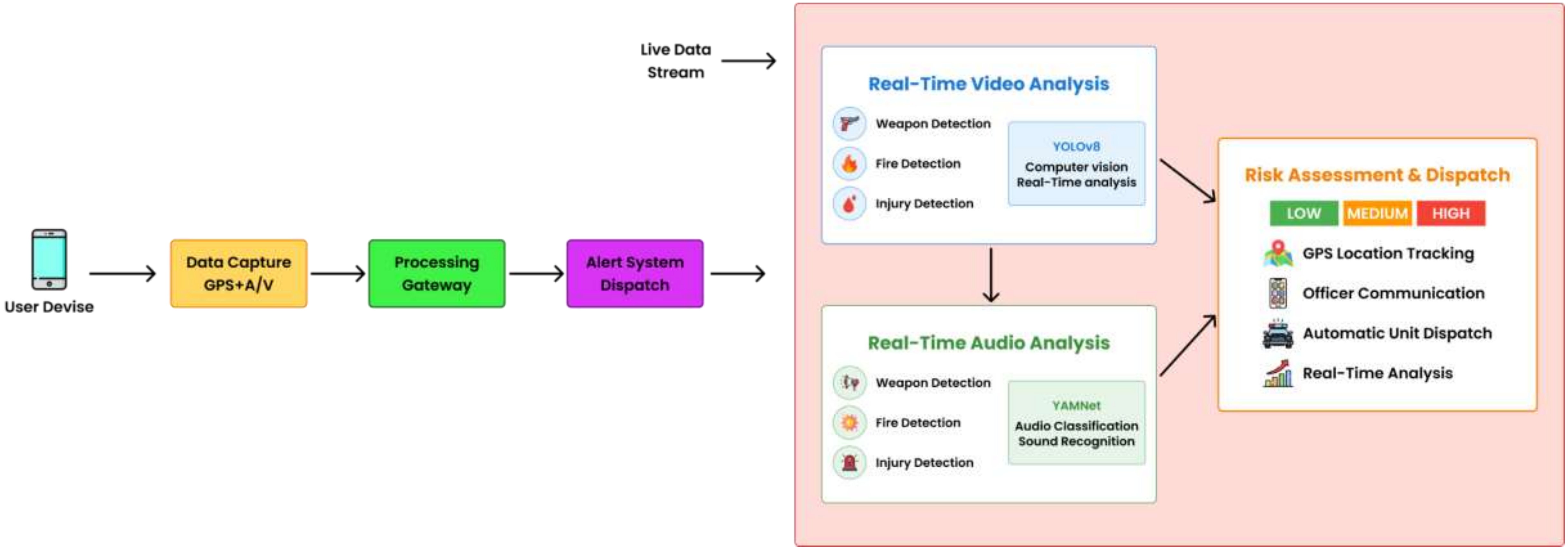
Emergency SOS with Real-Time Location & Live Streaming

Data Availability & Ethical Clearance

- **Data Sources:** Real-time GPS, video, and audio from mobile devices; public datasets for training (weapons, fire, screams, gunshots).
- **AI Models:** YOLOv8 (video) and YAMNet (audio) fine-tuned with open datasets.
- **Ethical Clearance:** User data encrypted, used only for SOS alerts, and shared securely with authorized responders.

Emergency SOS with Real-Time Location & Live Streaming

System Overview



Emergency SOS with Real-Time Location & Live Streaming

Key Domains


Artificial Intelligence


Machine Learning


Cloud Computing & Security


Real-Time Communication

Technologies

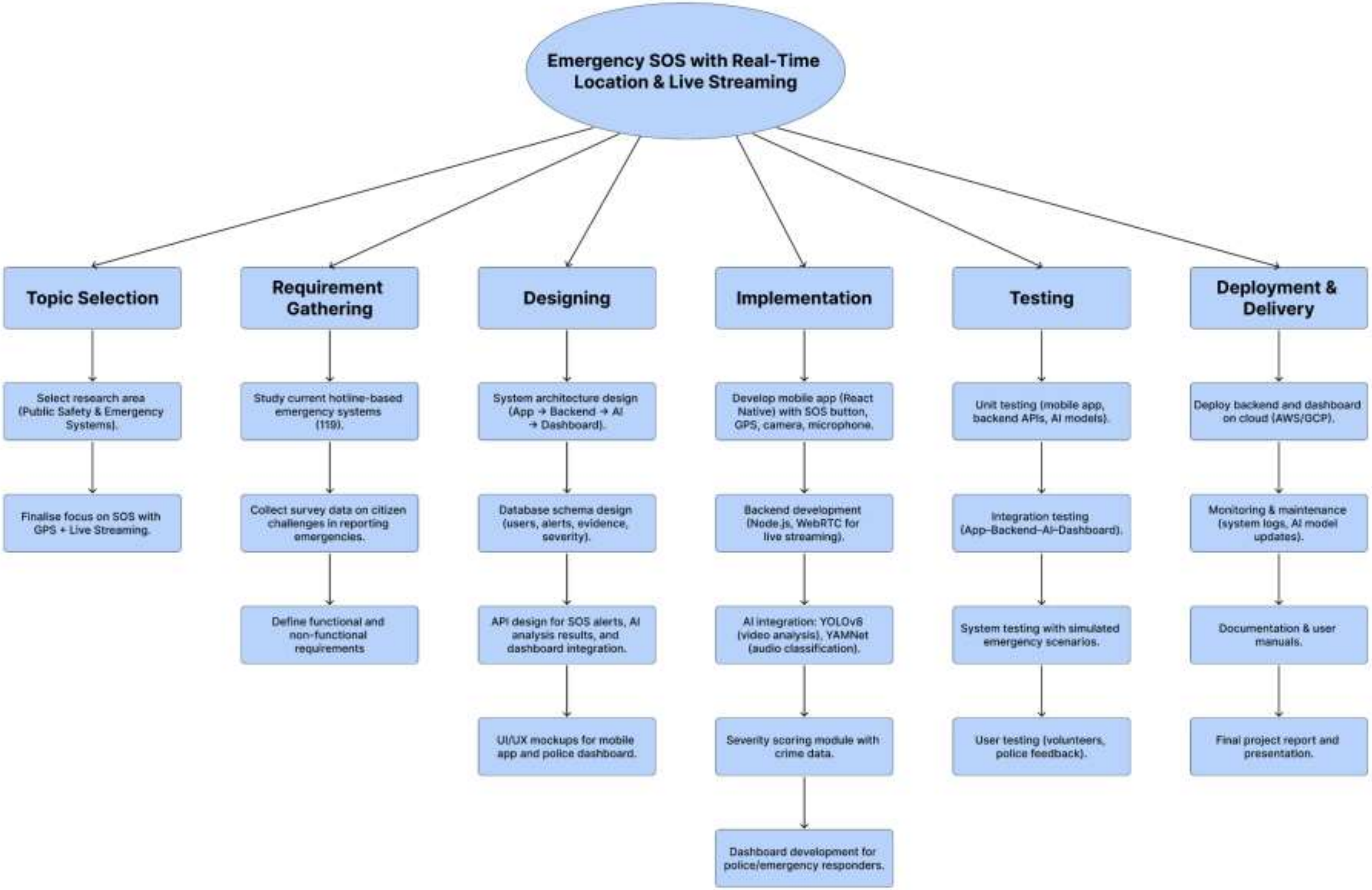
FastAPI

ultralytics YOLOv8

Firebase

yamnet

Work Breakdown Chart



References

[1] *SOS Emergency Alert and Assistance Mobile Application* (Ragavendran, 2024) – A mobile app that sends SOS alerts with GPS location and a text message to emergency contacts.

[2] *SOSerbia: Android-Based Software Platform for Sending Emergency Messages* (SOSerbia, 2018) – Android SOS app for sending emergency messages with basic location info.

[3] *Real-Time Weapon Detection Using YOLOv8 for Enhanced Safety* (Thakur et al., 2024) – Applies YOLOv8 for detecting weapons in live video streams in public spaces.

[4] *SOS Detection App – The Emergency Ally, 24/7* – Mobile system using sensor fusion (accelerometer, GPS, audio) to detect distress situations automatically.

REAL-TIME AI SYSTEM FOR DETECTING HARMFUL SOCIAL MEDIA CONTENT

Specialization : Information Technology

Real-Time AI System for Detecting Harmful Social Media Content

Knowledge Gap / Problem Definition

- Harmful content often appears on social media before reaching authorities.
- Manual or report-based monitoring cannot scale to volume, speed, and bilingual posts.
- Need a localized, real-time system that turns mixed-media signals into actionable alerts.

Shortcomings of Existing Systems

- No auto fetch /No real-time intake: Depend on manual uploads or slow batch scans.
- English-centric**:** weak handling of Sinhala, code-switching, and local slang.
- Single-modality blind spots**:** Text-only or video-only models miss cross-modal cues.

Real-Time AI System for Detecting Harmful Social Media Content

Research Gap

Features	R[1]	R[2]	R[3]	R[4]	Proposed Solution
Auto-Fetch from Social Media APIs	✗	✗	✗	✗	✓
Text Analysis (NLP)	✓	✓	✗	✗	✓
Video Analysis (CV)	✗	✗	✓	✓	✓
Audio Analysis (ASR)	✗	✗	✗	✓	✓
Multimodal Fusion & Decision	✗	✗	✗	✓	✓
Real-Time Alerts Dashboard	✗	✗	✗	✗	✓

Real-Time AI System for Detecting Harmful Social Media Content

Proposed Creative Solution

SafeLink

- Auto-fetch, not manual uploads
- Bilingual prefilter (SI + EN)
- Real-time alerts + evidence
- Categorize Posts / Prioritize

Real-Time AI System for Detecting Harmful Social Media Content

Key Domains

- Machine Learning
- Natural Language processing (NLP)
- Computer Vision (CN)
- Automatic Speech Recognition (ASR)
- Multi Model Fusion

Technologies

- Wav2Vec 2.0
- ultralytics YOLOv8
- BERT
- FastAPI
- TensorFlow
- Firebase

Real-Time AI System for Detecting Harmful Social Media Content

Evaluation & Validation

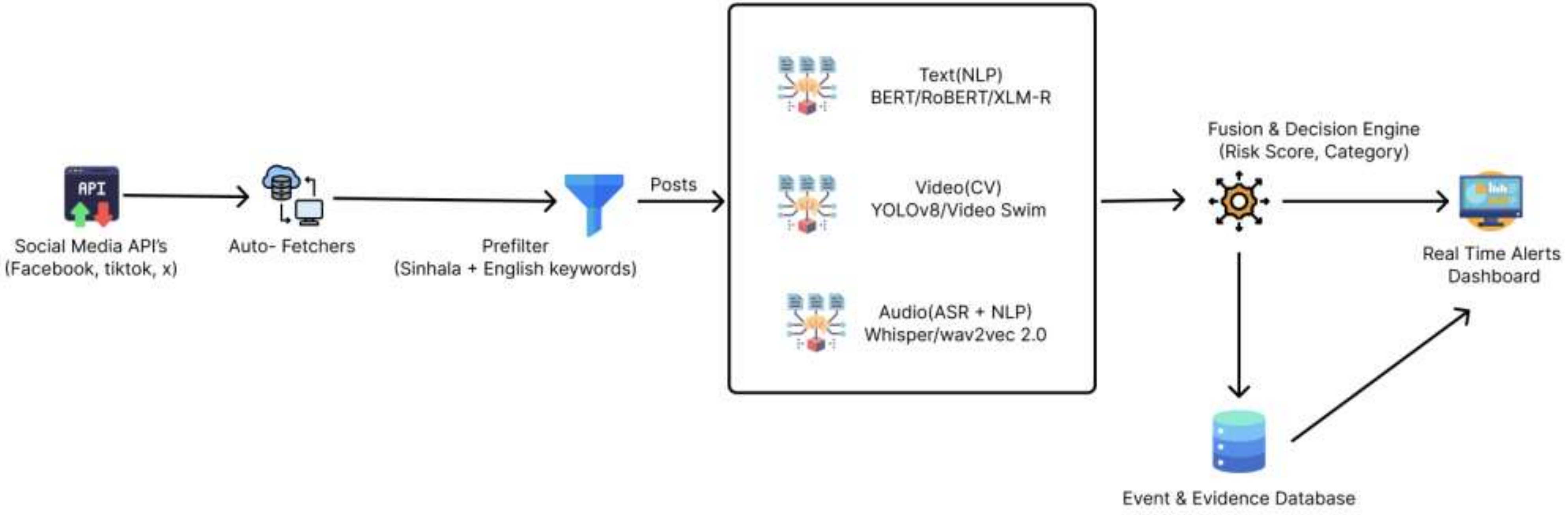
- Metrics: per-class **Precision, Recall, F1, PR-AUC**; **WER** for ASR.
- Ablations: single-modality vs multimodal, SI+EN vs EN-only; prefilter on/off.
- Stress tests for noisy audio, blur, slang, and reposts.

Data Availability & Ethical Clearance

- Social media **public endpoints** (Facebook Pages/Groups, YouTube channels, TikTok hashtags, X filtered streams).
- **Local news/public safety pages** for verified incident posts.
- **Open datasets** for pretraining Algorithms:
- Strictly follow **platform Terms of Service,** use official APIs where available.

Real-Time AI System for Detecting Harmful Social Media Content

System Overview

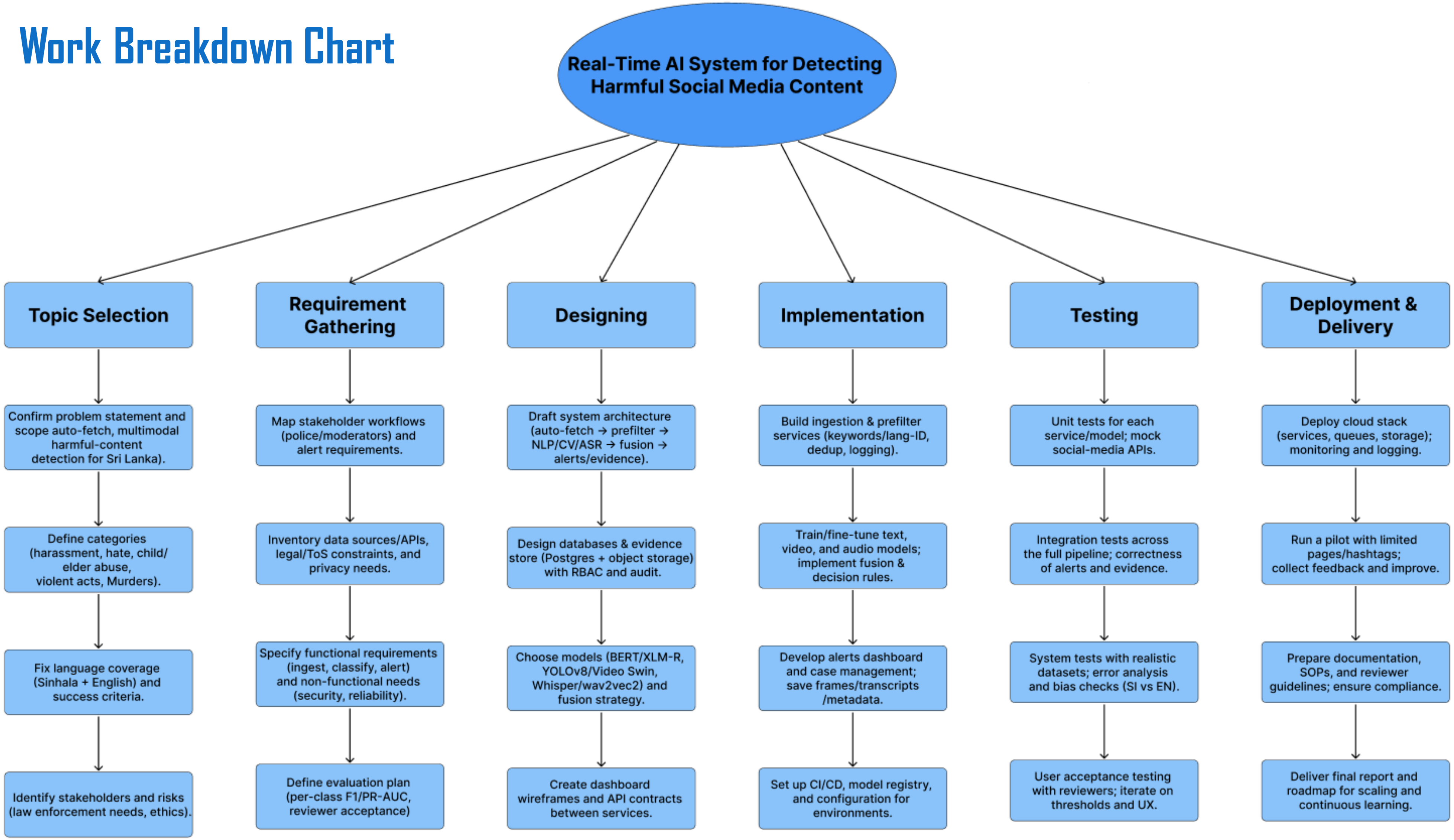


Real-World Application

- Faster intervention for violent/abusive incidents posted online.
- Actionable, evidence-backed alerts for police/moderators.
- Bilingual coverage tailored to Sri Lanka extensible to other public-safety uses.

Real-Time AI System for Detecting Harmful Social Media Content

Work Breakdown Chart



References

[1] U. Dikwatta and T. G. I. Fernando, "Violence Detection in Social Media—Review," *Vidyodaya Journal of Science*, vol. 22, no. 2, pp. 7–16, Dec. 2019.

[2] R. Pahujani, S. Bhuyan, S. Rai, and R. K. Kaliyar, "Abusive Content Detection Using Deep Learning: A BERT-Based Approach," in *Proc. 2024 IEEE 16th Int. Conf. on Computational Intelligence and Communication Networks (CICN)*, 2024, pp. 178–182.

[3] H. Alvari, S. Sarkar, and P. Shakarian, "Detection of Violent Extremists in Social Media," in *Proc. 2019 IEEE Conf. on Data Intelligence and Security (ICDIS)*, 2019, pp. 43–47.

[4] A. K. Saber, M. T. Gaata, and I. N. Ibraheem, "Violence Scene Detection from Social Media Networks Using Artificial Intelligence Methods," in *Proc. 2023 Int. Conf. on Information Technology, Applied Mathematics and Statistics (ICITAMS)*, 2023, pp. 178–182

SMART TRAFFIC VIOLATION DETECTION & ACCIDENT PREDICTION SYSTEM

Specialization : Information Technology

Smart Traffic Violation Detection & Accident Prediction System

SafeLink

Knowledge Gap / Problem Definition

- Road accidents remain a major cause of deaths in Sri Lanka. [1]
- Current enforcement methods. [2]
- Limitations of current systems. [3]
- Existing systems focus only on single tasks. [3]
- No integrated framework for real-time detection and prediction. [5]

Shortcomings of Existing Systems

- Violation detection works but without ALPR or accident prediction. [2]
- Accident hotspot analysis depends on offline crash data. [3]
- ALPR is unreliable in Sri Lankan conditions. [4]
- No integrated real-time framework for detection and prediction. [5]]

Smart Traffic Violation Detection & Accident Prediction System

Research Gap

Features	[1]	[2]	[3]	[4]	Safe Link
Real-time violation detection (speeding, red-light, overtaking)	✗	✓	✗	✗	✓
License plate recognition (ALPR) adapted for Sri Lankan plates	✗	✗	✗	✓	✓
Accident hotspot prediction with ML + clustering	✓	✗	✓	✗	✓
Citizen participation via mobile reporting	✗	✗	✗	✗	✓
Real-time alerts & police dashboard	✗	✗	✗	✗	✓
Integration of detection + ALPR + prediction in one framework	✗	✗	✗	✗	✓

Smart Traffic Violation Detection & Accident Prediction System

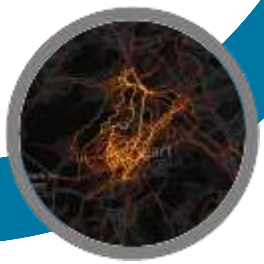
Proposed Creative Solution

SafeLink

» Real-time Traffic Violation Detection System

» Automated License Plate Recognition (ALPR) Module

» Intelligent Accident Prediction Models



» Citizen & Police Integrated Platform

Smart Traffic Violation Detection & Accident Prediction System

Key Domains

- Machine Learning
- Optical Character Recognition (OCR)
- Computer Vision (CV)
- Geospatial Analytics
- Mobile & Web Development

Technologies

ultralytics YOLO11

Firebase

FastAPI

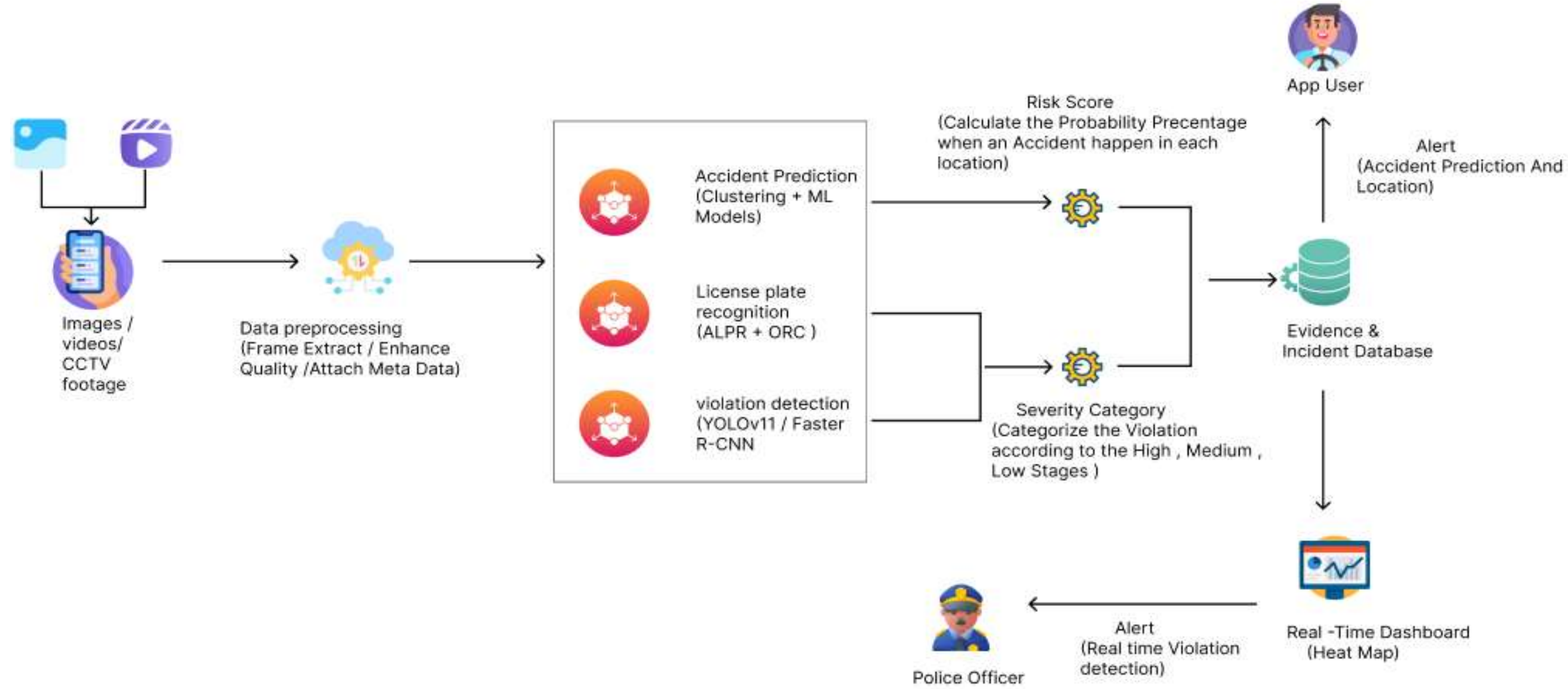
easyocr

React Native

Google Maps

Smart Traffic Violation Detection & Accident Prediction System

Methodology



Real-World Application

- Faster intervention for road accidents and traffic violations through real-time alerts.
- Actionable, evidence-backed reports with images, videos, and license plates for police.
- AI-powered hotspot prediction enables proactive deployment of traffic police.
- Citizen mobile app improves community participation and quick reporting.

Smart Traffic Violation Detection & Accident Prediction System

Evaluation & Validation

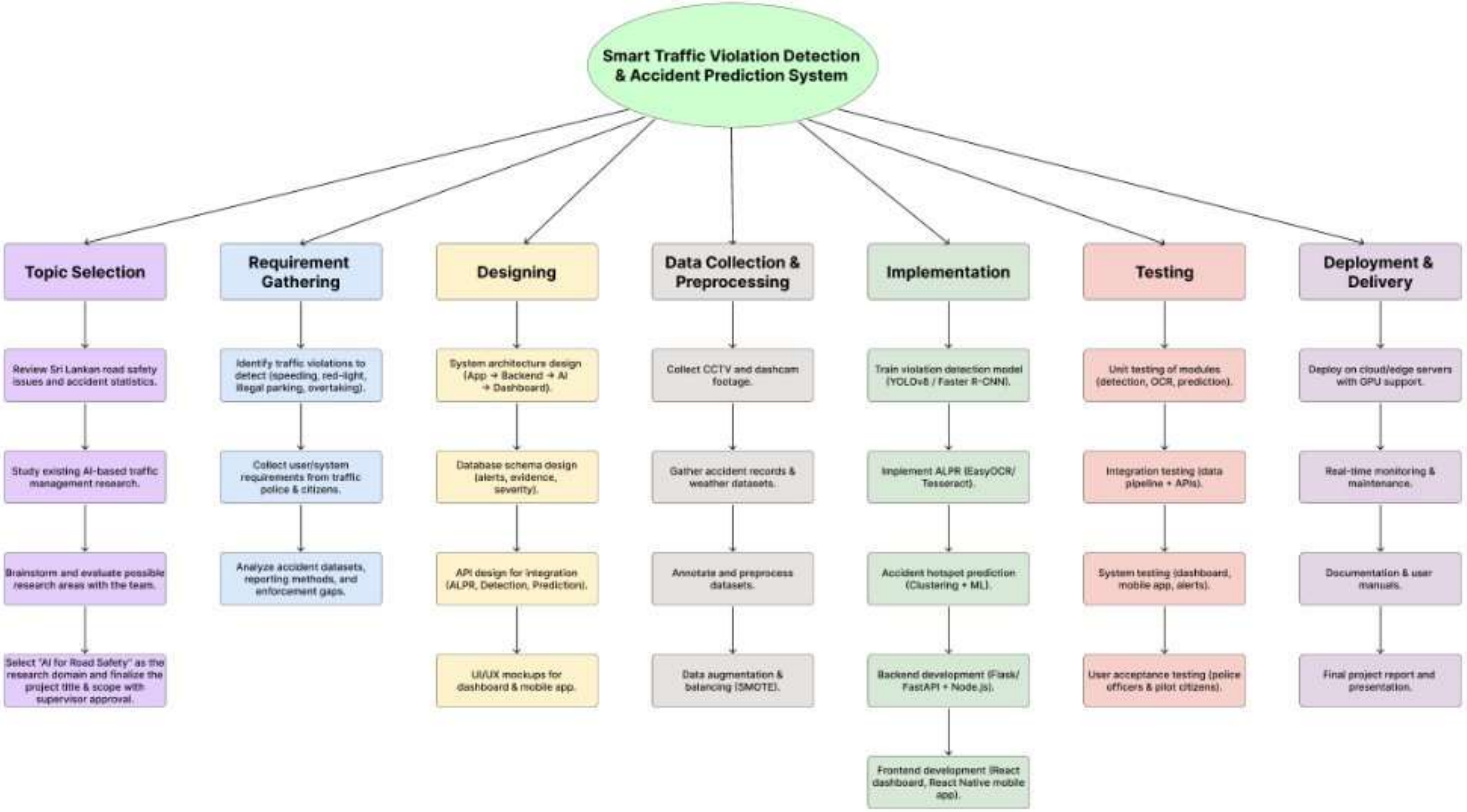
- Unit testing of AI modules (YOLOv11, ALPR, prediction).
- Integration testing of full pipeline.
- Metrics: Accuracy, Precision, Recall, F1, mAP.
- User testing with police & road users.

Data Availability & Ethical Clearance

- Sources: CCTV, dashcams, accident records, app reports.
- Data: Violations (speeding, red-light, parking, overtaking).
- Ethics: Anonymized data, encrypted storage, authorized access only.

Smart Traffic Violation Detection & Accident Prediction System

Work Breakdown Chart



References

[1] V. De Silva, T. M. Rengarasu, and J. P. S. S. Madushani, “Road traffic crashes and built environment analysis of crash hotspots based on local police data in Galle, Sri Lanka,” International Journal of Injury Control and Safety Promotion, 2018.

[2] A. Peiris, et al., “Computer Vision Based Approach for Traffic Violation Detection,” in Proc. Int. Research Conf., General Sir John Kotelawala Defense University (KDU), 2021.

[3] T. M. Rengarasu, et al., “Identification of traffic accident hotspots using network kernel density estimation,” Sri Lankan Journal of Applied Statistics, 2025.

[4] M. R. P. N. Bandara and X. Cheng, “An Automatic Number Plate Recognition System for Sri Lankan Vehicle Number Plates,” International Journal of Scientific Research and Engineering Development, vol. 6, no. 3, 2023.

[5] Asian Transport Observatory, “Sri Lanka Road Safety Profile 2025,” Asian Development Bank (ADB), 2025.

Thank You!